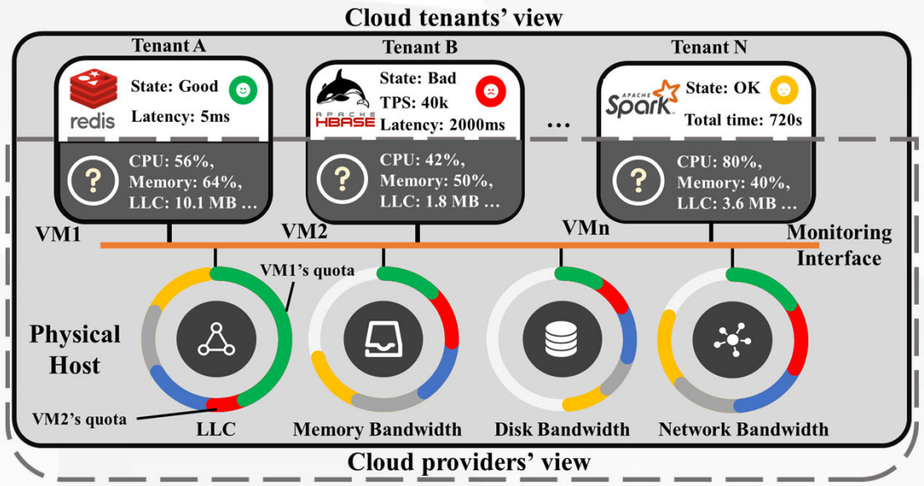
Research Summary of Tianyao Shi

- Research Interest:
  - My research interest focuses on Data Mining and Applied Machine Learning, especially for real-world applications, e.g., AI for Systems and Recommender Systems.
  - I wish to further explore robust and trustworthy ML methods and their applications in real-world data mining problems, e.g., learning on imbalanced and noisy data, OOD detection and domain generalization.
- Major Research Projects:
  1. QoS Prediction in Public Cloud – IPDPS 23 Publication
     (2021.01 – 2022.07, in collaboration with Huawei Cloud, onsite internship in Beijing during 2021.01-08)
  2. Predicting Enterprise Customers' IT Budgets for Cloud Services – WWW 24 Submission
     (2022.08 – 2023.10, in collaboration with Huawei Cloud, onsite internship in Hangzhou)
  3. Collaborative Recommender Systems – (2nd Author) SIGIR 21 and DASFAA 21 Publication
     (2019/12 – 2021/01)

Background & Motivation of Project 1

- In multi-tenant public cloud, multiple VMs contend for shared PM resources that cannot be perfectly partitioned, leading to deteriorated performance of cloud applications.



- Target:
  - Identify the VM suffering performance degradation caused by co-location interference via data-driven ML methods, how severe the degradation is, whether QoS violation occurs;
  - The inputs are features generated from system monitoring metrics (e.g., resource usage, hardware events), and the outputs are apps' relative performance loss.
- Challenges:
  1. Lacking off-the-shelf dataset,
  2. System application requires efficiency and interpretability,
  3. Must consider possible OOD samples in the inference stage.

Methodology of Project 1

- Framework Design Responding to Challenges

1. Lacking off-the-shelf dataset

   Solution: Conduct co-location experiments to mimic real-world interference in cloud and collect metrics

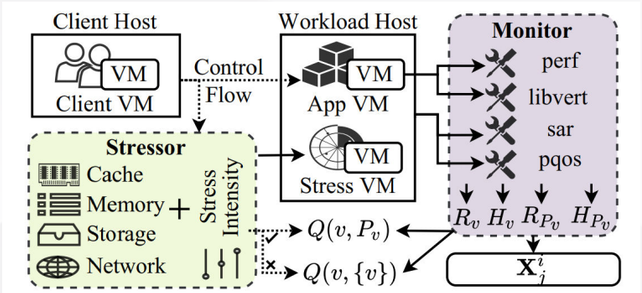
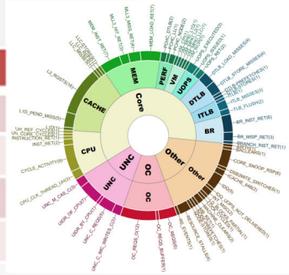
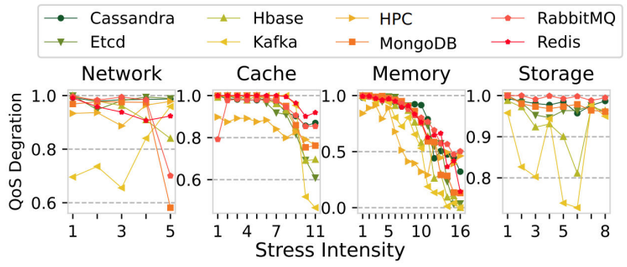


TABLE I: Hardware Specification

CPU	Intel Xeon Gold 6151
Virtualization technology	QEMU-KVM + Openstack
Host OS	Red Hat 2.8.5 (kernel 3.10)
Guest OS	Ubuntu 16.04 (kernel 4.14)
Cores, Sockets	2 sockets, 18 physical cores per socket
Core frequency	fixed at 3.0GHz, DVFS disabled
Hyperthreading	Enabled, 2 threads per physical core
Last-level (L3) cache	22.5MB, 11 ways

App. Type	App. Selected
HPC	SPEC CPU 2006 Suite
NoSQL	YCSB-{hbase, cassandra, mongodb}
Message Middleware	Kafka, RabbitMQ
K-V Store	Redis, Etcd



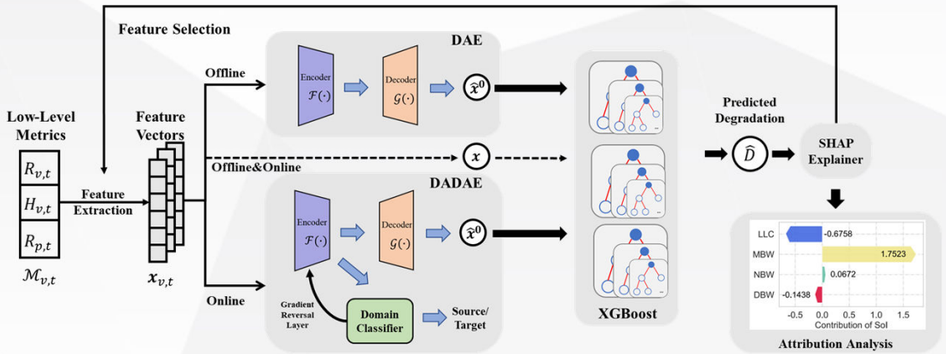


2. Must consider possible OOD samples in the inference stage
3. System application requires efficiency and interpretability

   Solutions:

   2 -> Adding online domain adaptation module

   3 -> Performing feature selection to reduce #of metrics monitored; adding a explainer based on shapely values and doing a case study



Conclusion & Lessons of Project 1

Conclusions

- Observations on data characteristics is vital
  - When there is interference, the values of low-level metrics can be seen as a combination of interference noise and no-interference values.
  - So use a DAE to augment the input data by estimating metrics under no interference.
- Building dataset is hard, data scarcity is challenging
  - Learned a ton of system & arch knowledge to build the dataset
  - System guys relunctunt to data-driven methods (better after GPT's success)
  - DM/ML/AI guys avoid problems without off-the-self dataset
- Every design must have its rationales
- Real-world applications calls on robust and trustworthy AI

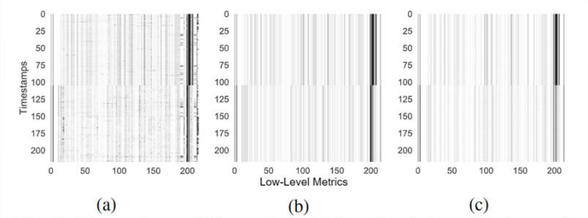


Fig. 5: Comparison of Cassandra's (a) interfered data, (b) data under no interference and (c) the interfered data processed by DAE. The horizontal axis represents 218 low-level metrics. The vertical axis represents 220 consecutive timestamps.

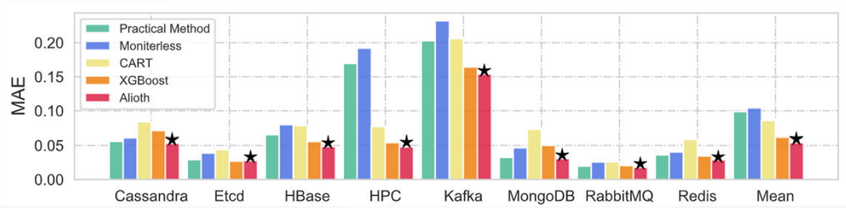
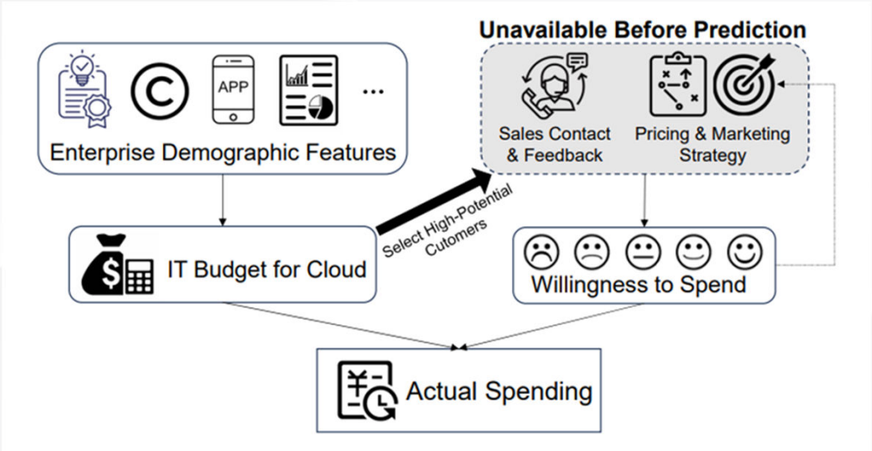


TABLE III: Online MAE of Practical Method, Moniterless, CART, XGBoost, and Alioth on different applications.

	Cassandra	Etcd	Hbase	HPC	Kafka	MongoDB	RabbitMQ	Redis	Mean
Homogeneous Train&Test	0.0508	0.0277	0.0470	0.0542	0.1481	0.0372	0.0207	0.0375	0.0529
Oracle DAE+XGBoost	0.0820	0.0654	0.0550	0.1264	0.1720	0.0426	0.0707	0.0567	0.0838
Practical Method	0.1323	0.7844	0.0677	0.1723	**0.2124**	0.2877	0.1963	0.1587	0.2514
Monitorless	0.1287	0.5673	0.0891	0.2197	0.3144	0.2503	0.1538	0.1479	0.2339
CART	0.1264	0.9602	0.0743	0.5741	0.4442	0.0997	0.1776	0.1373	0.3242
XGBoost	0.1102	0.2658	**0.0545**	0.2453	0.3037	0.0784	0.1432	0.1440	0.1681
Offline Alioth	0.1376	1.5446	0.0883	0.7635	0.295	0.3713	0.226	0.2512	0.4597
Online Alioth	**0.1047**	**0.0839**	0.0650	**0.1628**	0.2373	**0.0487**	**0.0782**	**0.0847**	**0.1082**

Background & Motivation of Project 2

- Cloud platforms mainly rely on enterprise customers. It's lucrative to estimate IT budgets for cloud services of yet-to-adopt-cloud enterprises, s.t. we can pick out high-potential customers and launch targeted marketing campaigns to acquire new customers.



- Target:
  - The inputs are static demographic features generated from enterprise info (e.g., #of patents, registered capital), and the outputs are IT budget values.
- Why not just predict spending?
  - Budget and Willingness are two independent latent factors that co-decide spending;
  - Willingness can be arbitrary and is unpredictable: no information implying willingness as the enterprises haven't adopted cloud;
  - Cannot acquire willingness unless the sales persons make actual contact.
  - A very special new problem with real-world demands.

Challenges & Method of Project 2

- Framework Design Responding to Challenges

1. No ground-truth label for budgets: we only have spending records and they are different

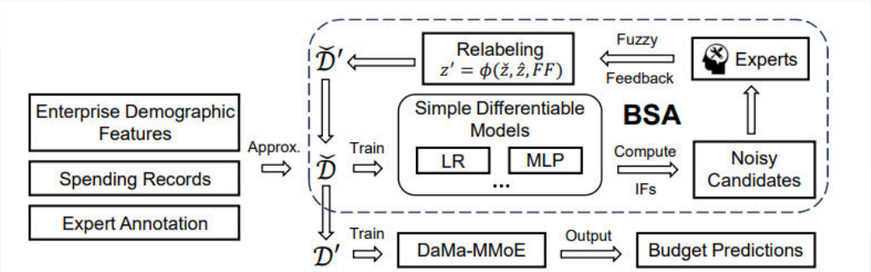
   Solution: Approximate budgets with spending records and view the approximation errors as label noise; Detect noisy labels via Influence functions and relabel them in a human-in-the-loop manner.

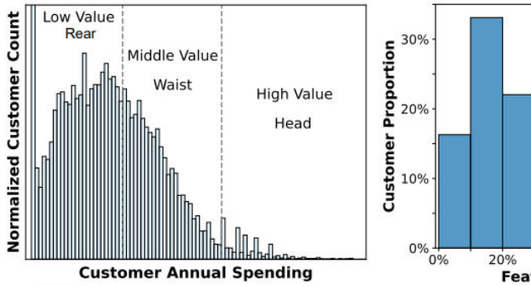
2. Sales team demands both fine-grained prediction (i.e., regression) on high-value customers and coarse-grained prediction (i.e., classification) on others, coupled with long-tail label distribution

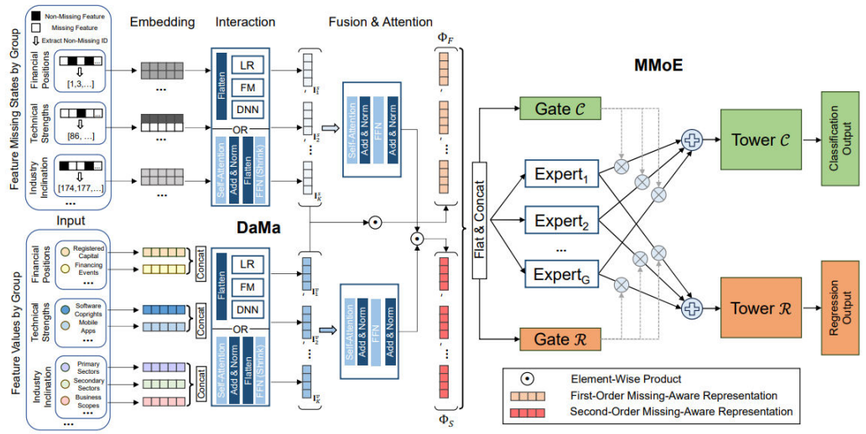
   Solution: Do multi-tasking; Use MMoE to alleviate the scarcity in head customers & learn task-specific representations

3. Real-world data has very high feature missing ratios

   Solution: Porpose a Dual-attention Missing-aware (DaMa) Network to encode feature-missing states and learn intra- and inter-group missing-aware representations







Conclusion & Lessons of Project 2

Conclusions

- Each component's design is validated through experiments
  - However, the problem is new and special and the baseline LTV prediction methods aren't designed for this problem, which may not be fair.
- Industrial production methods and fancy paper SOTAs are two worlds; Good offline validation results do not guarantee useable online inference results
- Real-world data is vast of quality issues
  - Class imbalance / rare categories, label noise, data scarcity …
  - Robust ML is once again called on for being widely deployed
- Submitted a WWW 24 paper under review.

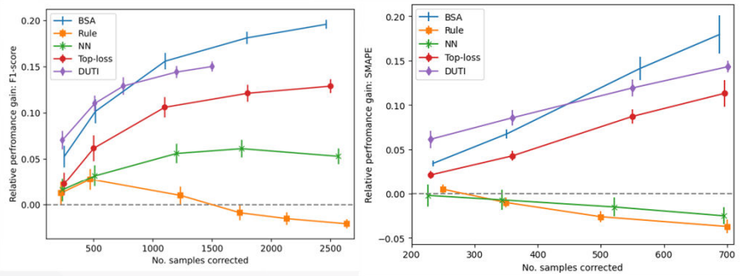


Table 2: Performance of different methods on two tasks.

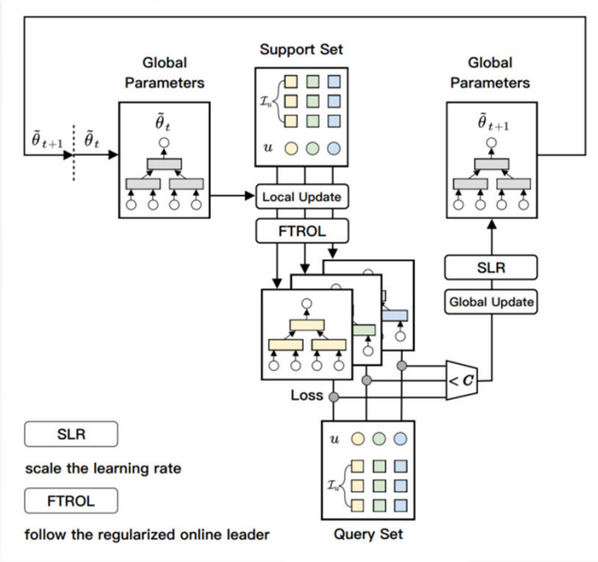
Methods	F1	AUC	SMAPE	N.Gini
XGBoost	0.4667	0.7164	0.3471	0.5528
ZILN	0.3968	0.6332	0.3273	0.6052
MDME	0.5349	0.8101	0.2905	0.6824
Bidden-MarfNet	0.4120	0.6712	0.2820	0.6917
DaMa-MMoE	**0.5488**	**0.8239**	**0.2746**	**0.7011**

Table 3: Performance of different variants of DaMa-MMoE.

Variants	F1	AUC	SMAPE	N.Gini
OFMR	0.5369	0.8122	0.2845	0.6803
OSMR	0.5411	0.8163	0.2794	0.6912
NMI	0.5337	0.8077	0.2954	0.6728
TSM	**0.5533**	**0.8291**	0.3117	0.6216
Shared-Bottom	0.5481	0.8212	0.2986	0.6786
DaMa-MMoE	0.5488	0.8239	**0.2746**	**0.7011**

Project 3

- FROM: Following the Online Regularized Meta-Leader for Cold-Start Recommendation (SIGIR21)
  - Existing meta-learning based recommender systems alleviate the cold-start problem and achieve SOTA in offline validation
  - Real-world RS favor online solution with special challenges
    - Users present a long-tailed distribution, and more experience should be borrowed from user-level models with abundant historical records
    - Online system is highly volatile during training as the support set users are randomly sampled from noisy sequences
    - Large daily throughput of meta-learning based methods challenges the computing capacity of online systems
  - A novel following the online regularized meta-leader method is proposed to learn online gradients and resist noise efficiently.
  - I was only responsible for some parts of writing & visualization work, the first author got the most work done.



Algorithm 3: Follow the Regularized Online Leader

Input: Hyperparameters $\alpha, \tau, \lambda_1, \lambda_2$, Dataset D

Initialize $\mathbf{Z} = \{z_1, \cdots, z_N\} = \mathbf{0} \in \mathbb{R}^N$, and $\mathbf{Q} = \{q_1, \cdots, q_n\} = \mathbf{0} \in \mathbb{R}^N$;

$G = \{g_1, \cdots, g_N\} = \nabla_{\theta_t} \mathcal{L}_t(\boldsymbol{\theta}_t, D)$;

for $i \leftarrow 1, 2, \cdots, N$ **do**

  $\sigma_i \leftarrow \frac{1}{\alpha}\sqrt{q_i + g_i^2} - \sqrt{q_i}$;

  $z_i \leftarrow z_i + g_i - \sigma_i \theta_i$, $q_i = q_i + g_i^2$;

  if $|z_i| \geq \lambda_1$ **then**

    $\theta_{t,i} \leftarrow -(\lambda_2 + \frac{\tau + \sqrt{q_i}}{\alpha})^{-1}(z_i - \lambda_1 \mathrm{sgn}(z_i))$;

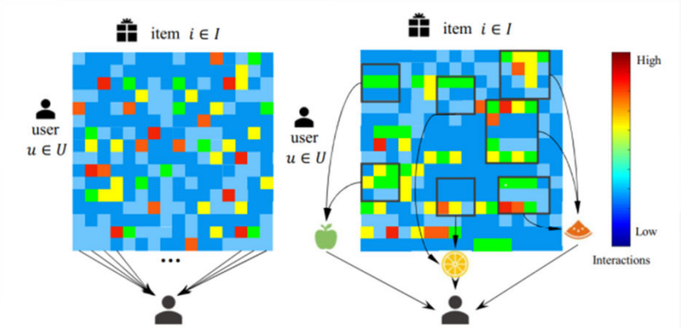
  else

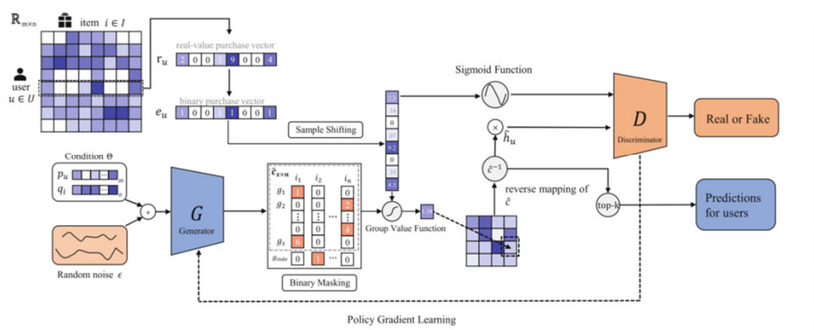
    $\theta_{t,i} \leftarrow 0$;

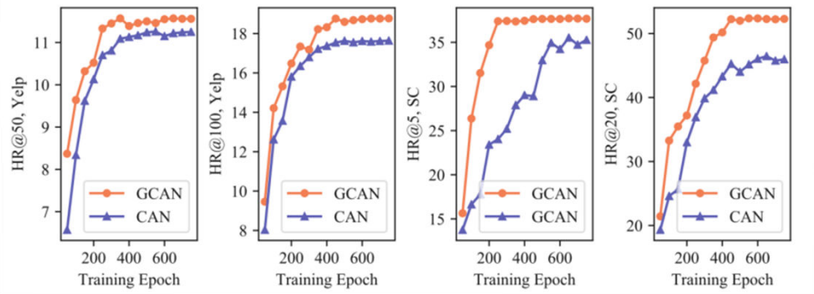
Return $\boldsymbol{\theta}_t$

Project 3

- GCAN: A Group-Wise Collaborative Adversarial Networks for Item Recommendation (DASFAA21)
  - GAN-based recommender systems were popular at that time, but are hard to converge and suffer from data sparsity
  - Item-based group-wise mechanism is introduced to enhance Collaborative Adversarial Networks based on data observations
  - Two novel tricks are proposed:
    - *Binary masking* abstracts group-wise relationship from binary user-item interaction
    - *Sample shifting* adds a small purtubation to mapping vectors to avoid learning all-zero vectors and promote the convergence process
  - I was only responsible for some parts of writing & visualization work, the first author got the most work done.







Thanks for listening!

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